Hugo Ruivo

Senior Software Engineer | CGI Pipelines & Production Systems

Portugal | English and Portuguese

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Professional Summary

Senior software engineer and Real-Time R&D Lead with 10+ years across automotive CGI, real-time 3D, and simulation. Designs connected production systems spanning vehicle data, artist-facing tools, Python services, Unreal Engine, and rendering infrastructure. Leads collaborative engineering work from architecture through delivery. Personal projects explore compilers, telemetry, RISC-V, and FPGA design.

Core Skills

Languages: C/C++, Python, Zig, Rust, TypeScript, C#

Pipeline & Production Engineering: CGI pipelines, automated rendering, Deadline, Perforce, custom asset ingestion, data normalization, pipeline tooling, CI/CD

Web & Backend: FastAPI, REST APIs, SvelteKit, Bun, Docker

Real-Time & DCC: Unreal Engine, Unity, Houdini, Blender, Omniverse

Infrastructure & Automation: Jenkins, Git, Perforce, DCC scripting and automation, Unreal Engine automated testing

Software Engineering: Modular architecture, data-driven systems, performance optimization, cross-functional development

Professional Experience

Real-Time R&D Lead / Pipeline Software Engineer | Digital Giant

May 2022 – Present

- Architected a multi-brand automotive CGI pipeline integrating Unreal Engine production tooling, Python services, web interfaces, source control, CI/CD, and automated rendering.
- Designed data-ingestion and conversion workflows for brand-specific vehicle assets, normalizing heterogeneous input data into a common structure consumed by downstream pipeline tools.
- Developed artist-facing tooling and rendering workflows integrating Unreal Engine with a FastAPI REST backend and SvelteKit/Bun web applications, enabling render jobs to be initiated from Unreal or a browser-based interface.
- Designed and implemented Perforce stream architecture for shared vehicle data, reusable Unreal Engine plugins, and project-specific development workflows across automotive brands.
- Built and maintained production automation infrastructure, including Jenkins CI/CD workflows, Docker-based deployment of web services, and custom Deadline render plugins.
- Developed Unreal Engine rigging and production tools while leading technical architecture and engineering delivery across artists and developers.
- Extended and customized existing Unreal Engine tools to boost artist workflows

CGI Real-Time R&D Lead | Hogarth Worldwide

2020 - 2022

- Led development of an Unreal Engine vehicle configurator with multiple environments and cinematic sequences.
- Integrated web-based systems with Unreal Engine for product validation and stakeholder feedback.

Technical Artist | Burrows

2019 - 2020

- Created Unreal Engine tools for vehicle import, configuration, and real-time rendering workflows.
- Built a WebGL vehicle configurator with an automated Houdini pipeline to optimize complex assets for browser rendering.

Software Developer | Mackevision

2018 - 2019

- Maintained Unreal Engine vehicle configurators using Pixel Streaming and extended interactive functionality.
- Implemented communication between an iPad control interface and the main Unreal Engine application.

Software Developer | Abyssal (now Ocean Infinity)

2016 - 2018

- Developed an Unreal Engine remotely operated vehicle simulator for the oil and gas industry.
- Implemented buoyancy, tether physics, hydraulic manipulator simulation, multi-display rendering, and custom hardware input integration.
- Designed modular vehicle systems, data-driven missions, and session recording; led developers and artists as the prototype expanded.

Game Programmer | X-Team

2012 - 2016

- Developed web and mobile game projects using Unity.

Selected Personal Projects

AI Assisted Car Configurator

Integrated AI agents with an Unreal Engine vehicle configurator using Python, C++, and OpenAI/llama models.

Telemetry and Mission Control

Telion telemetry backend and a web dashboard for monitoring KSP/DCS vehicles and sending commands. Built an adapter to display NASA Artemis II telemetry using the same data format and dashboard.

Education

Game Development Engineering | Polytechnic Institute of Cavado and Ave

Barcelos, Portugal | 2009 - 2012